Page 1: Spring Boot Core Architecture and Dependency Injection

Spring Boot simplifies enterprise application development through convention over configuration. Dependency Injection allows the IoC container to wire collaborators together at runtime. Constructor injection is preferred over field injection because it ensures immutability, guarantees complete object instantiation, and makes unit testing straightforward without reflection.

Page 2: Java 21 Virtual Threads and Concurrency

Project Loom introduced Virtual Threads in Java 21 as lightweight implementations of java.lang.Thread.
Virtual threads are scheduled by the JVM onto carrier platform threads rather than operating system threa
When a virtual thread blocks on blocking I/O like database socket reads or REST calls,
the JVM unmounts it from the carrier thread, allowing other virtual threads to execute.
This dramatically increases throughput for high-concurrency I/O-bound web services.

Page 3: PostgreSQL Vector Indexing and Semantic Search

pgvector enables vector similarity search directly within PostgreSQL relational databases.

The HNSW (Hierarchical Navigable Small World) index creates a multi-layered graph for approximate nea

HNSW provides high recall and low search latency without needing an offline training step like IVFFlat.

Cosine similarity measures the orientation between vectors rather than their magnitudes.